Data Science and Artificial Intelligence

Machine Learning

Multi Layer Perceptron

Lecture No. 01

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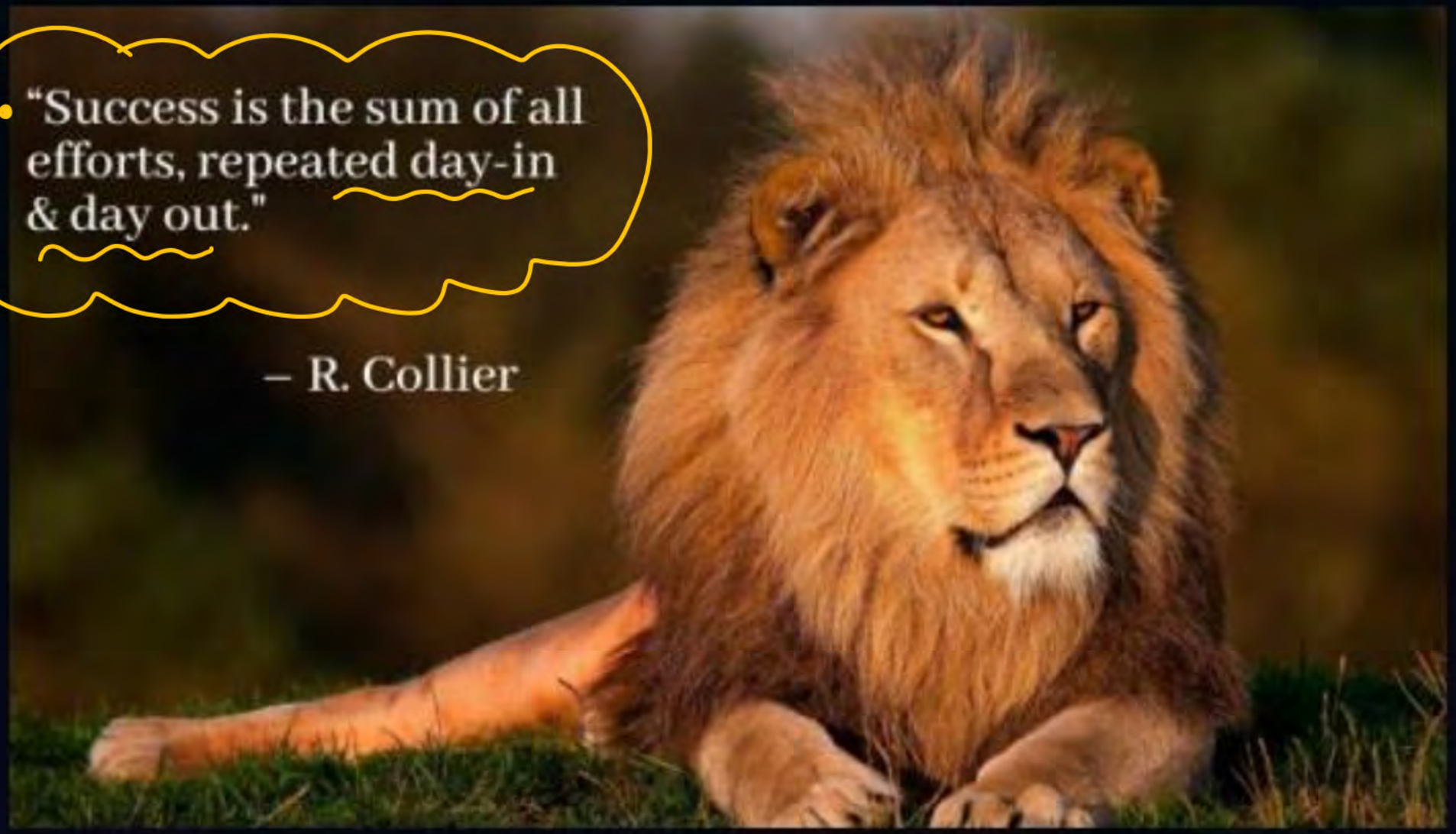
Recap of Previous Lecture

- Topic — PCA and LDA
- Topic
- Topic
- Topic
- Topic

Topics to be Covered

- Topic
- Topic
- Topic
- Topic
- Topic

Perceptron learning
↓
Single & Multi layer
↓
Neural Network.


"Success is the sum of all efforts, repeated day-in & day out."
– R. Collier

Perceptron learning

What is Perceptron learining

- Perceptron is a building block of an Artificial Neural Network
- Perceptron is a linear Machine Learning algorithm used for supervised learning for various binary classifiers

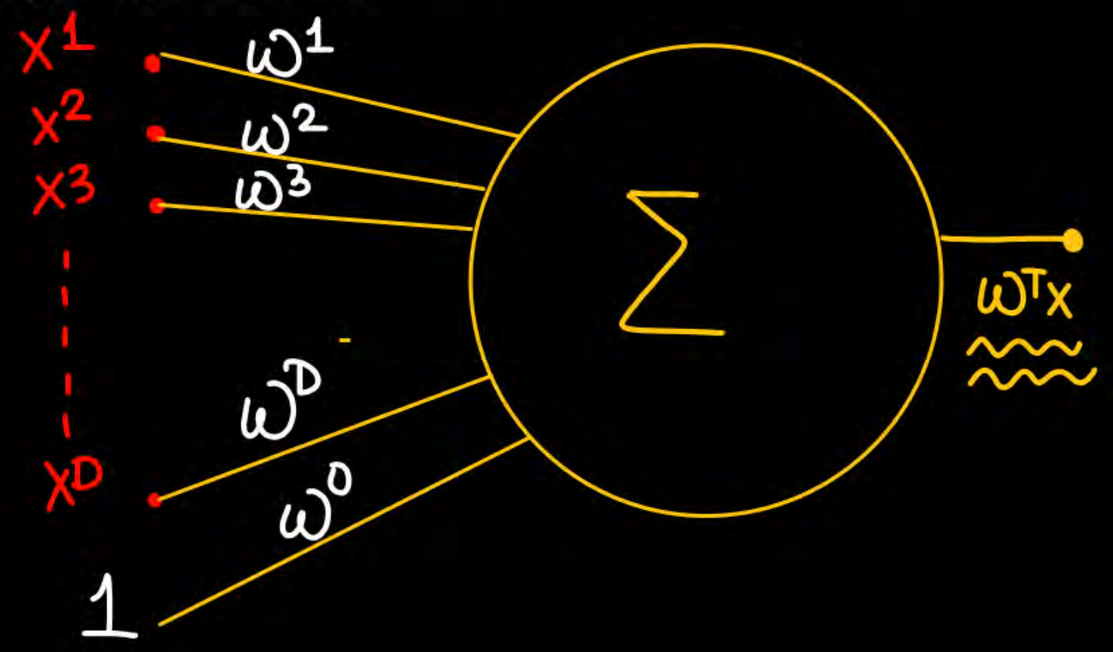
→ linear ML algo
→ binary classification

• It is a Single cell in NN

Perceptron learning

What is Perceptron learining

- We can consider it as a single-layer neural network with four main parameters, i.e., input values, weights and Bias, net sum, and an activation function.



$(w^T x + b)$

→ Remove b here

$(w^T x)$, $w = (w_0\ w_1 \ldots w_D)$

$x = (1, x^1, x^2 \ldots x^D)$

Perceptron learning

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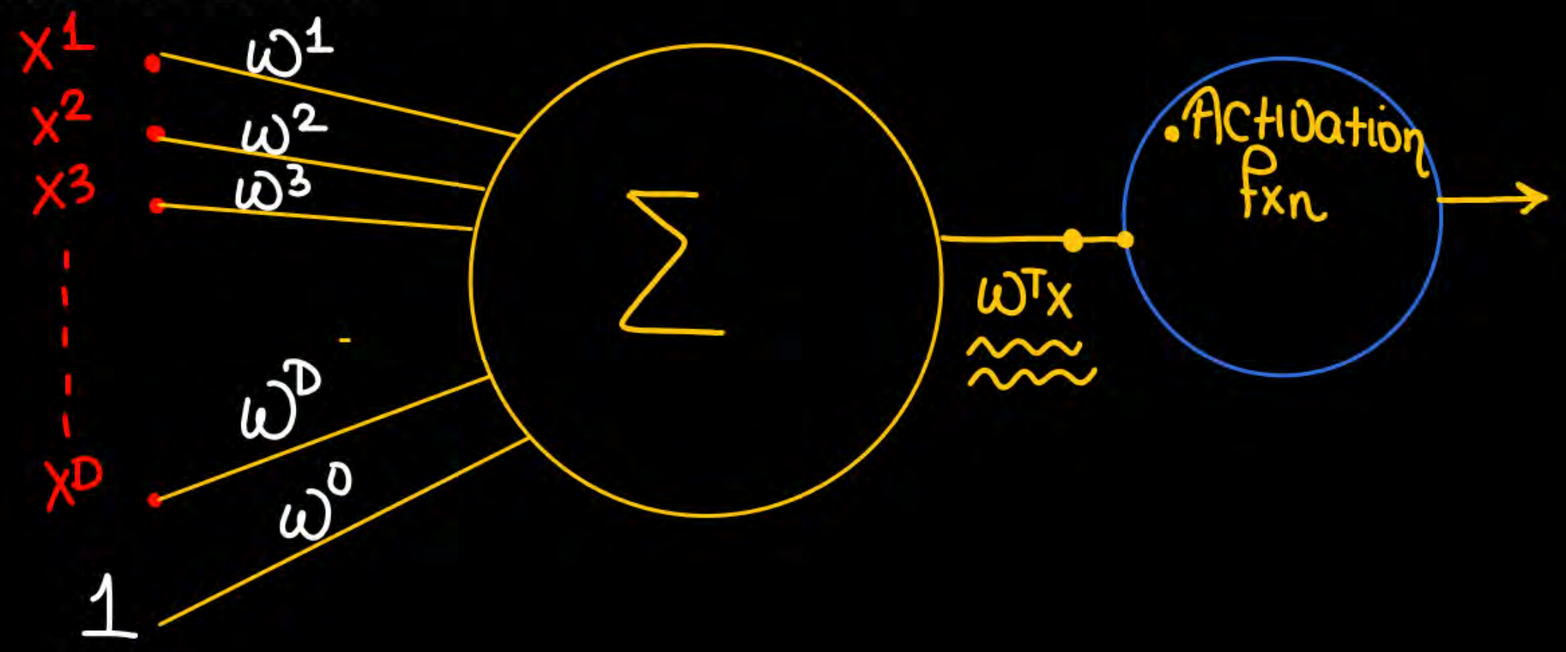
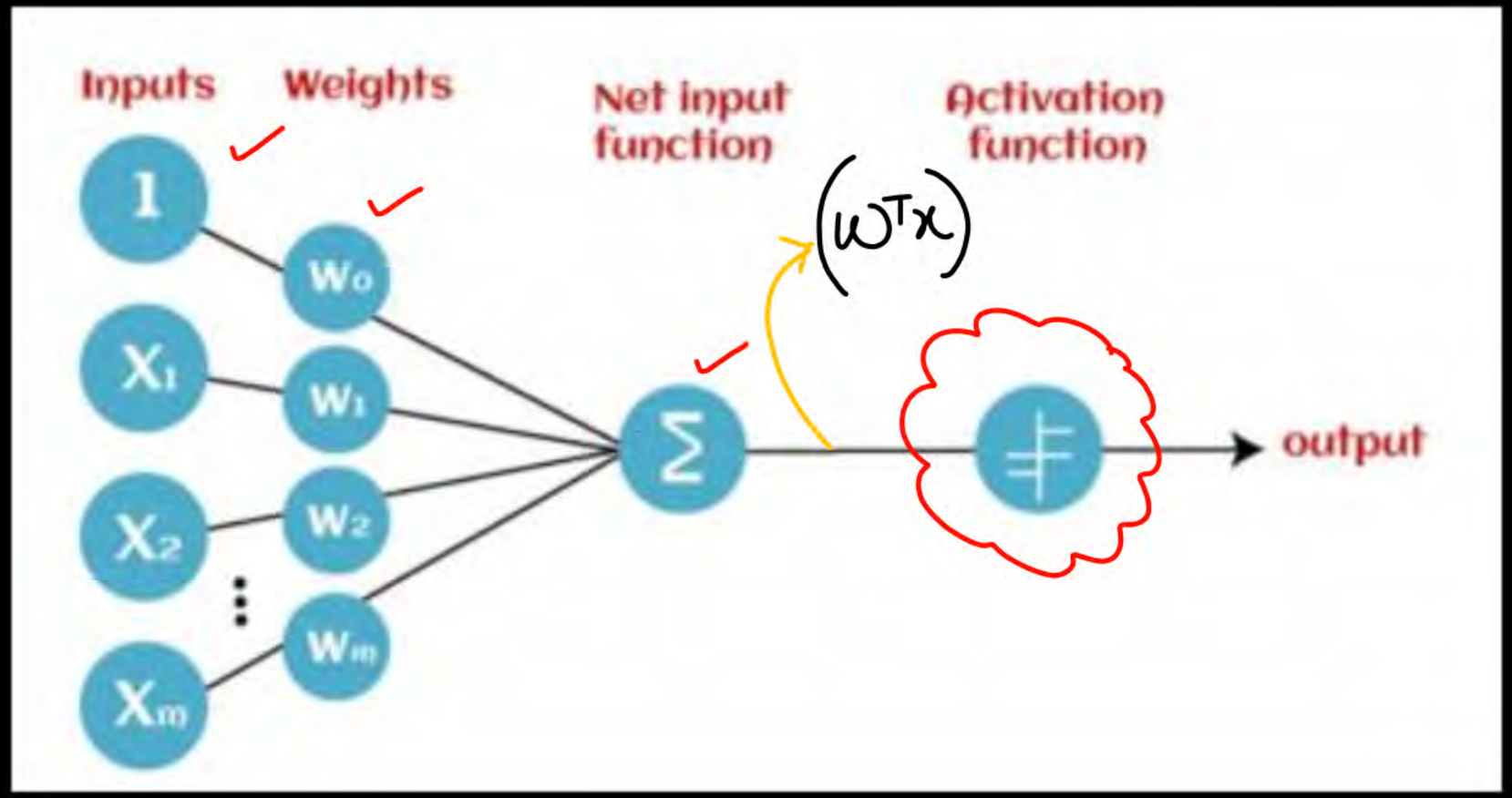


Perceptron learning

What is Perceptron learining



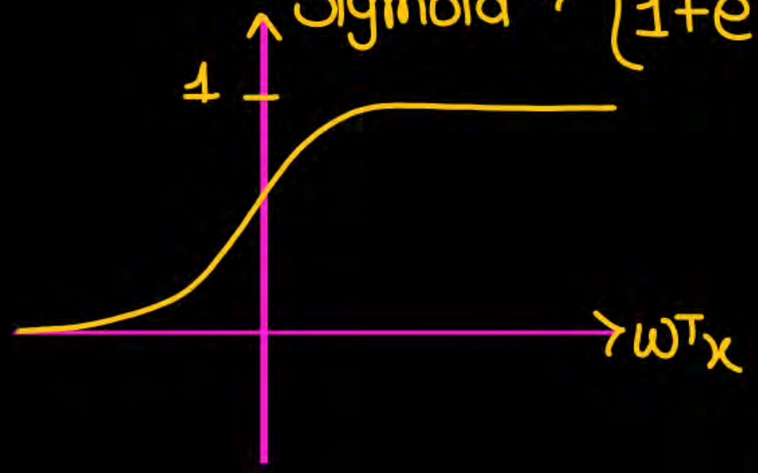
(• So Today we will learn the training of the Cell
• we have to determine the weights)

Perceptron learning

What is Perceptron learining

3 activation fxn.

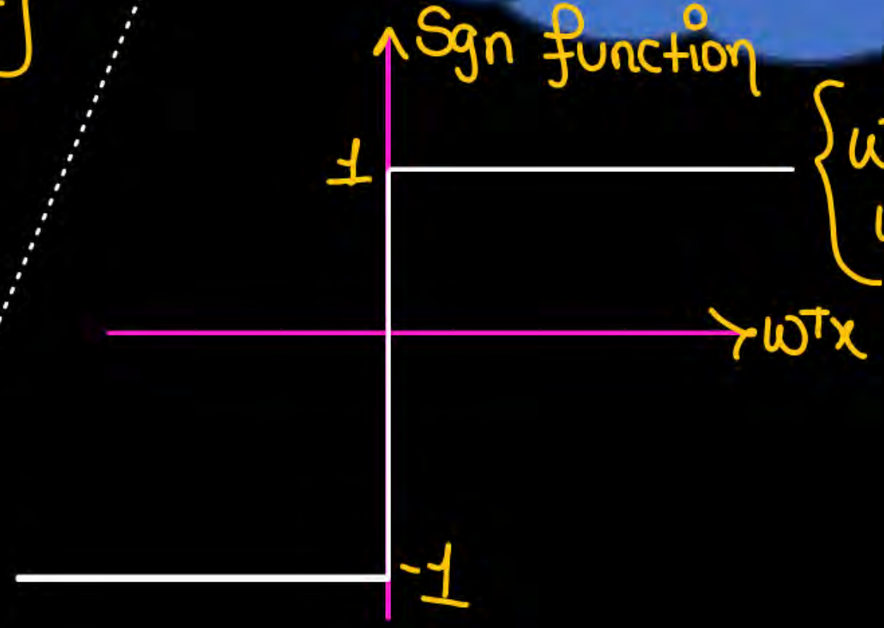
Sigmoid $\Rightarrow \left\{\frac{1}{1+e^{-w^T x}}\right\}$



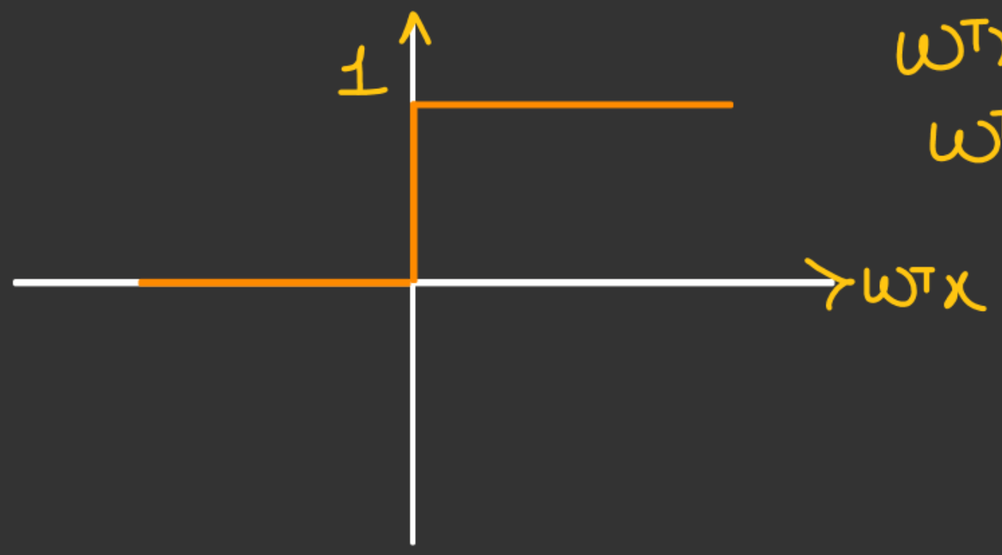
We can have 3 activation function...

Sgn function

$\left\{\begin{array}{l} w^T x > 0 \Rightarrow 1 \\ w^T x < 0 \Rightarrow -1 \end{array}\right\}$



Step fxn



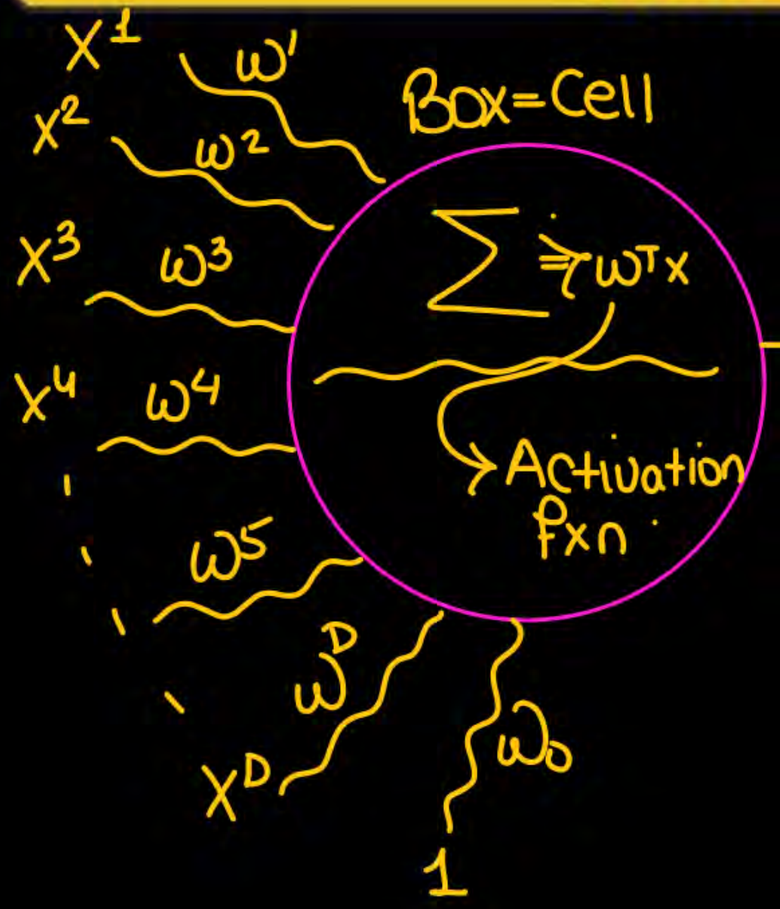
$w^T x > 0 \Rightarrow 1$

$w^T x < 0 \Rightarrow 0$

Perceptron learning

What is Perceptron learining

What is a discriminant function



@ O/P it produce Class of point

- inside a single cell $\Rightarrow$ $w^T x$ is Calculated and this is given to activation fxn that Produce O/P Result

Perceptron learning

What is Perceptron learining

What is a linear discriminant function

- So the discriminant fxn is and fxn that take in input X_i and produce a result that help in determining class of X_i
- if $g(x)$ is a discriminant fxn

$$\begin{cases} \text{if } g(x) > 0 \Rightarrow \text{Class 1} \\ g(x) < 0 \Rightarrow \text{" } 0 \end{cases}$$

we have a linear discriminant fxn

$\Rightarrow g(x) = w^T x$

$$\left\{ \begin{array}{l} w^T x > \theta \Rightarrow \text{Class } 1 \\ w^T x < 0 \Rightarrow \text{Class } 0 \end{array} \right\}$$

$\Rightarrow$ activation fxn [Sgn fxn] uses $w^T x$

So how to find w ?

⇒ So min error on training data

⇒ Error on training data

we have some w, how to find that classifier is commiting error on a point

$y_i(w^T x_i) < 0$

So we have 4 cases

$y_i = -1$, $w^T x_i < 0$ → no error	$y_i = +1$, $w^T x_i > 0$ →
$y_i = -1$, $w^T x_i > 0$ → error, Type 2 error	$y_i = +1$, $w^T x_i < 0$ → error, Type 1 error

So we will try to minimize

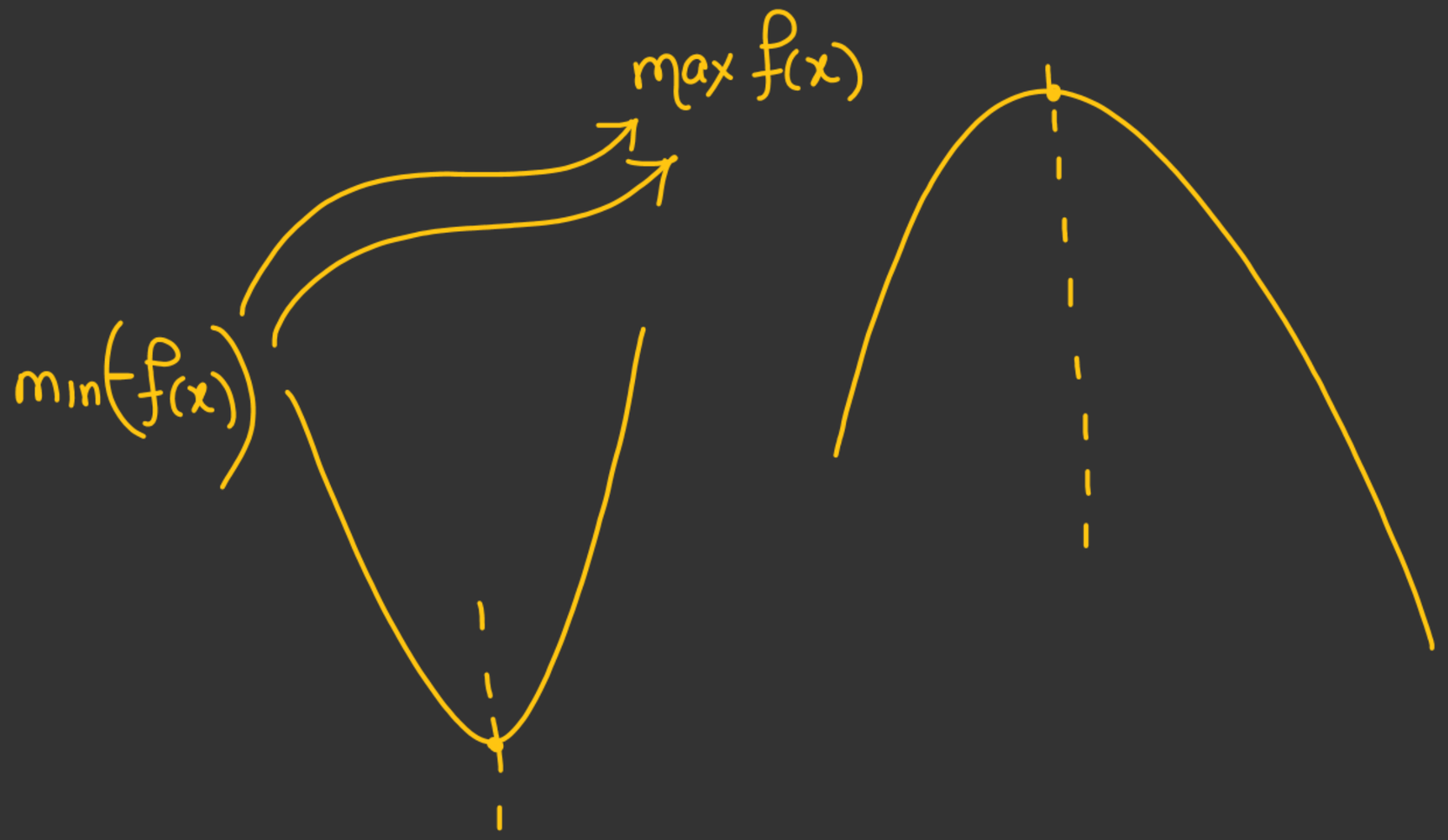
$$\Rightarrow \cancel{\sum_i y_i(w^T x_i)}$$

$$\min \quad \Rightarrow -\sum_i y_i(w^T x_i)$$

- So if M is the set of misclassified points then we try to minimize

$$-\sum_{i=1}^{M} y_i(w^T x_i)$$

$\hookrightarrow$ for all misclassified point


max f(x)
min(-f(x))

How to find w's

we start with any w

now if any point is correctly classified then w is not change

but if any point is misclassified then we try to minimize $\underbrace{-y_i^{\circ}(w^T x_i^{\circ})}_{E}$

{gradient descent} So now update w

$$w^{new} = \left(w^{old} - \eta \frac{\partial E}{\partial w}\right) = \left[w^{old} + \eta\, y_i x_i^{\circ}\right]$$

$$E = -y_i(w_0 + w_1 x_i^1 + w_2 x_i^2 \cdots w_D x_i^D)$$

$$\frac{\partial E}{\partial w} = \begin{bmatrix} \partial E/\partial w_0 \\ \partial E/\partial w_1 \\ \vdots \\ \partial E/\partial w_D \end{bmatrix} = \begin{bmatrix} -y_i \\ -y_i x_i^1 \\ -y_i x_i^2 \\ \vdots \\ -y_i x_i^D \end{bmatrix}$$

$$\frac{\partial E}{\partial w} = \begin{bmatrix} \partial E/\partial w_0 \\ \partial E/\partial w_1 \\ \vdots \\ \partial E/\partial w_D \end{bmatrix}$$

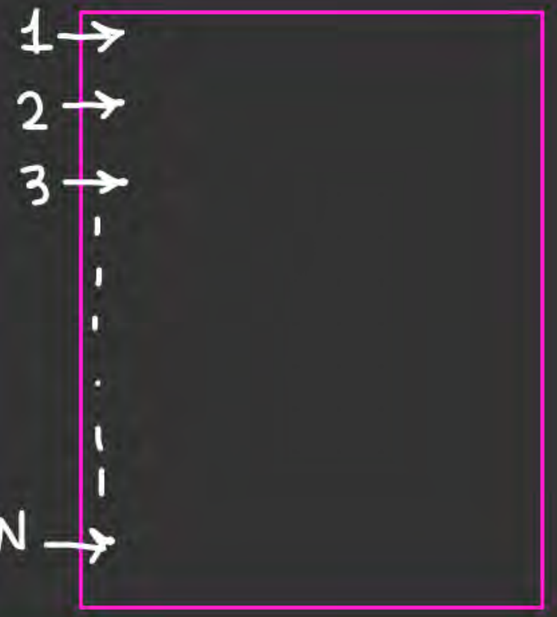
Actual algo

1. Start with w's $\Rightarrow$ all zero
2. data point's Padded with Row of all 1's
3. Initialize $m=0$
4. Now Start testing the training data one by one
5. If the point has $y_i(w^T x_i) > 0 \Rightarrow$ No need to update w move ahead
6. If the point has $y_i(w^T x_i) \leq 0 \Rightarrow$ error

$$w^{t+1} = (w^t + \eta y_i x_i)$$

$m = m+1$

η : learning Rate



5, 6 step Run for all points

So after Running 5th 6th step's N times

One iteration is completed.

7. Now check, m, if $m \neq 0$ then our ω is not good

go back to step No 3, with last updated ω,

This algo will Run until we get 'ω' which give zero error.

Perceptron learning

What is Perceptron learining

How to Find W's

```
Initialize w⃗ = 0⃗
while TRUE do
    m = 0
    for (x_i, y_i) ∈ D do
        if y_i(w⃗^T · x⃗_i) ≤ 0 then
            w⃗ ← w⃗ + y x⃗
            m ← m + 1
        end if
    end for
    if m = 0 then
        break
    end if
end while
```

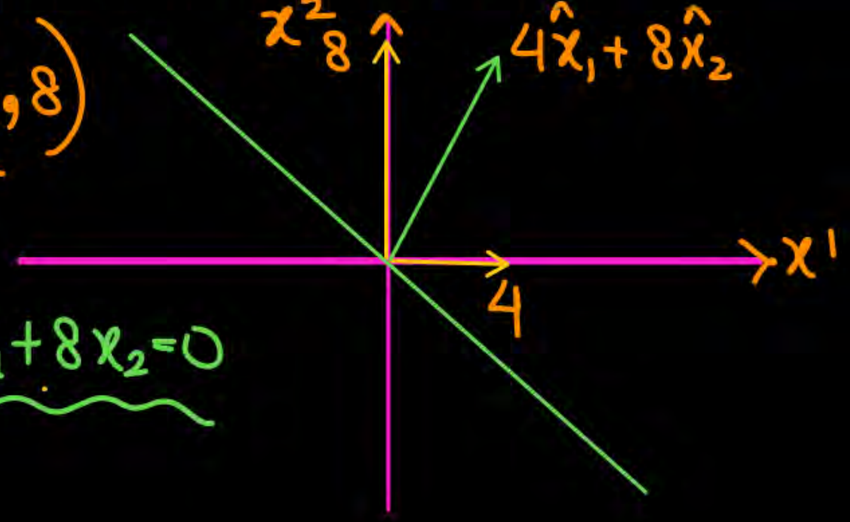
Perceptron learning

Lets see an example

Lets see geometrically

So if we have a vector in 2D

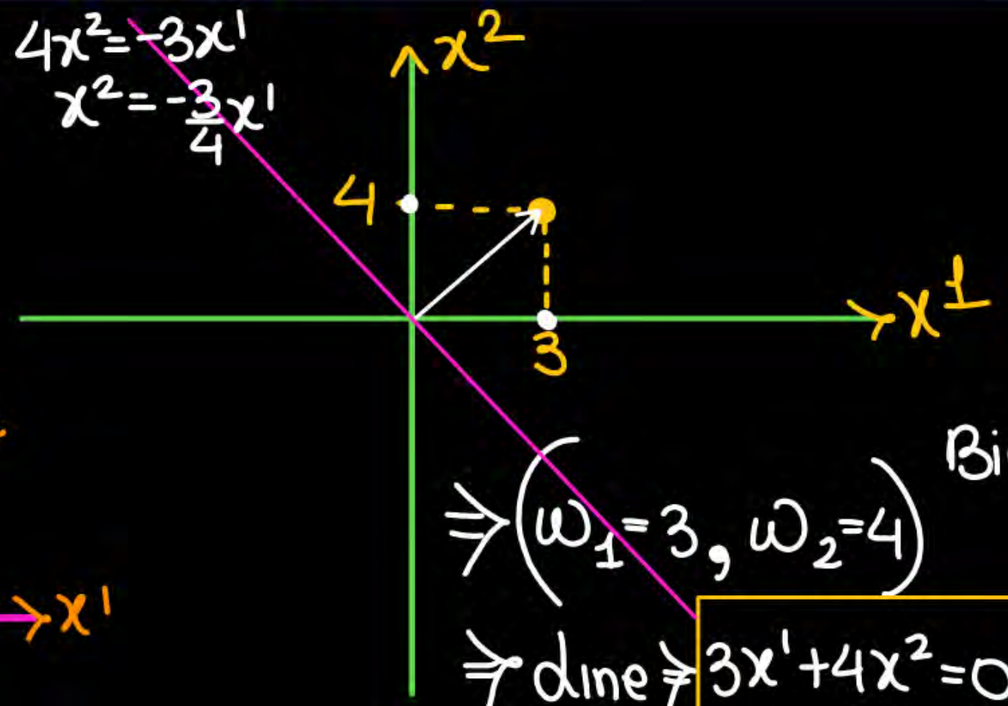
$\Rightarrow (4, 8)$



$\Rightarrow 4x_1 + 8x_2 = 0$

$4x^2 = -3x^1$

$x^2 = -\frac{3}{4}x^1$



Bias=0

$\Rightarrow (w_1 = 3, w_2 = 4)$

$\Rightarrow$ line $\Rightarrow$ $\boxed{3x^1 + 4x^2 = 0}$

Perceptron learning

(Bias=0)

Lets see an example

$\Rightarrow w's = 0$

$\Rightarrow y_i(w^T x_i) = 0$ error

$\Rightarrow w^{new} = w^{old} + \eta y_i x_i$

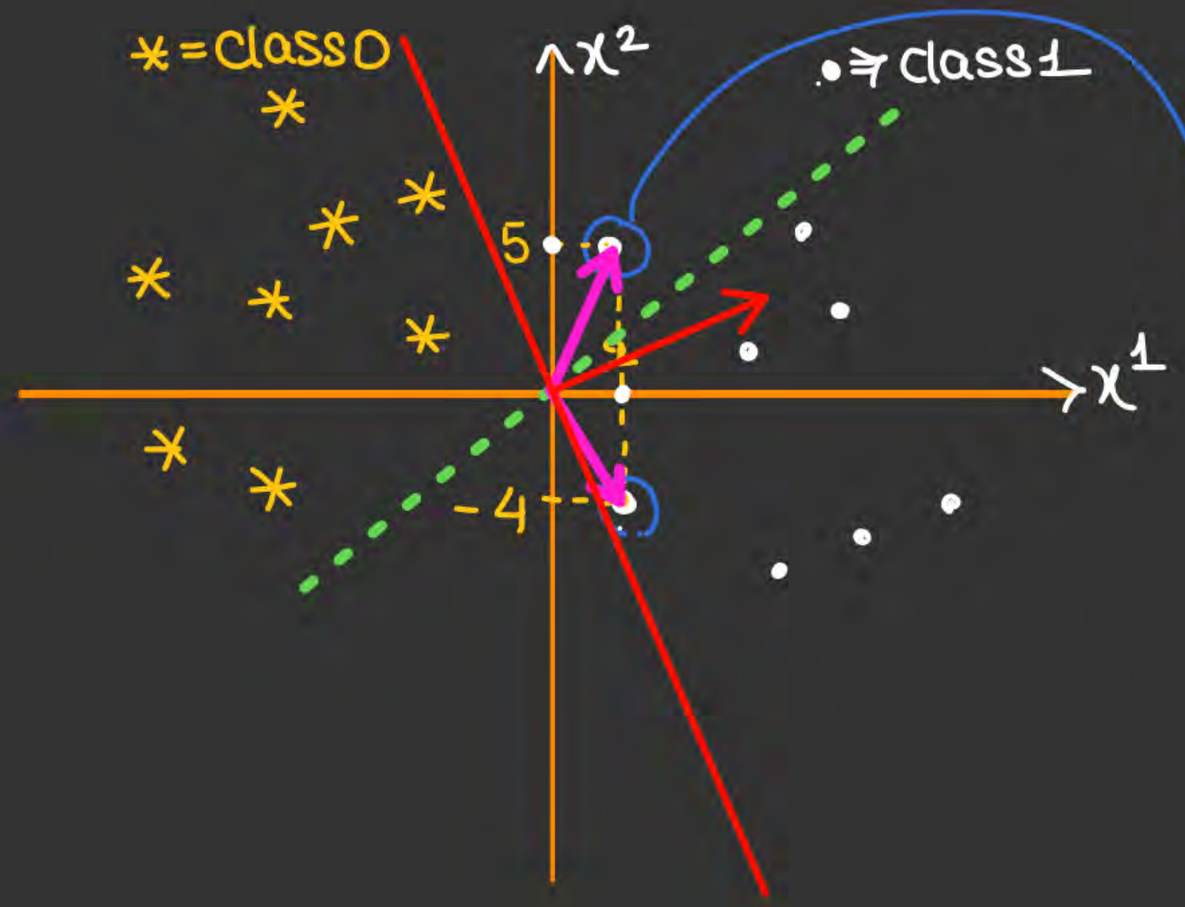
$\eta = 1$

$\Rightarrow w^{new} = 0 + 1.1.x_i$

$w^{new} = x_i$

$w^{new} = (2,-4) \Rightarrow 2\hat{x}_1 - 4\hat{x}_2$

(Bias=0)



$w = (2, -4)$

$\Rightarrow y_i(w^T x_i) < 0$ error

$w^{new} = w^{old} + \eta y_i x_i$

→ error

$$w^{new} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} + 1 \cdot 1 \cdot x_i$$

$$= \begin{bmatrix} 2 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

Perceptron learning

Lets see an example

So we need 2 iteration

* In this algo if we get Perfect 'w' in 5th iteration

So the algo is such that we need 1 more iteration

Perceptron learning

Lets see an example

Input $\mathbf{x}$	Output y
(2, 3)	1
(1, 1)	1
(4, 5)	0
(3, 2)	0

$$w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

	1st	2nd	y
1)	1	2, 3	$\to 1$
2)	1	1 1	$\to 1$
3)	1	4 5	$\to -1$
4)	1	3 2	$\to -1$

Start $w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ $\quad \eta = 1$

$\to$ Point No 1 $\Rightarrow y_i(w^T x_i) = 0$ error

$$w^{new} = w^{old} + \eta y_i x_i = 0 + 1 \cdot 1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$w = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\to$ Point No 2 $\quad y_i(w^T x_i) \Rightarrow y_i(w_0 + w_1 x^1 + w_2 x^2)$

$\Rightarrow 1(1 + 2 \times 1 + 3 \times 1) \Rightarrow +ve$

$\Rightarrow$ no error

$\to$ Point No 3 $\Rightarrow y_i(w^T x_i) = -1(1 + 2 \times 4 + 3 \times 5) < 0$

$$w^{new} = w^{old} + \eta y_i x_i \Rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}$$

Point No 4 $\Rightarrow y_i(w^T x_i) \Rightarrow -1(0 - 2 \times 3 - 2 \times 2)$

$\Rightarrow +ve$

No error

$$\Rightarrow w = \begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix}$$

#Q. You are a Hollywood producer. You have a script in your hand and you want to make a movie. Before starting, however, you want to predict if the film you want to make will rake in huge profits, or utterly fail at the box office. You hire two critics A and B to read the script and rate it on a scale of 1 to 5 (assume only integer scores). Each critic reads it independently and announces their verdict. Of course, the critics might be biased and/or not perfect, therefore you may not be able to simply average their scores. Instead, you decide to use a perceptron to classify your data. There are three features: a constant bias, and the two reviewer scores. Thus $f_0 = 1$ (a constant bias), f_1 = score given by reviewer A, and f_2 = score given by reviewer B.

#Q. Train the perceptron to generate hat $\hat{Y} = 1$ if the movie returns a profit, hat $\hat{Y} = -1$ otherwise. The initial weights are $\omega_0 = -1$, $\omega_1 = 0$, $\omega_2 = 0$ Present each row of the table as a training token, and update the perceptron weights before moving on to the next row. Use a learning rate of $\alpha = 1$ After each of the training examples has been presented once (one epoch), what are the weights?

Movie Name	A	B	Profit
Pellet Power	1	1	No
Ghosts!	3	2	Yes
Pac is bac	4	5	No
Not a Pizza	3	4	Yes
Endless Maze	2	3	Yes

HW

#Q. Suppose that, instead of learning whether or not the movie is profitable, you want to learn a perceptron that will always output hat $\hat{Y} = 1$ when the total of the two reviewer scores is more than 8, and hat $\hat{Y} = -1$ otherwise. Is this possible? If so, what arethe weights ω_0, ω_1, and ω_2 that will make this possible?

#Q. Instead of either part (a) or part (b), suppose you want to learn a perceptron that will always output hat $\hat{Y} = +1$ when the two reviewers agree (when their scores are exactly the same), and will output hat $\hat{Y} = -1$ otherwise. Is this possible? If so, what are the weights ω_0, ω_1, and ω_2 that will make this possible?

THANK - YOU